Suspected Stroke

UP-14 | Universal Patient Care

Quick flow

01 Recognize Stroke [EMT / AEMT / Paramedic]

Cincinnati screen • posterior-circulation warning signs • exact last known well

02 Complete C-STAT [EMT / AEMT / Paramedic]

Gaze 2 • LOC questions + commands 1 • arm weakness 1 • score ≥2 suggests LVO

03 Glucose + Mimics [EMT / AEMT / Paramedic]

Check glucose • treat hypoglycemia under UP-4 • repeat neurologic examination

04 Protect Brain + Airway [EMT / AEMT / Paramedic]

NPO • manage secretions • protect affected side • oxygen only for hypoxemia

05 Stroke Alert + Destination [EMT / AEMT / Paramedic]

Follow current Claiborne Stroke Destination Plan • consider C-STAT and transport time

06 Monitor + Access [AEMT / Paramedic]

Cardiac monitor • 12-lead if no delay • one IV preferred • NS TKO

07 Rapid Transport [EMT / AEMT / Paramedic]

Scene target ≤15 minutes • serial neurologic exams • transmit stroke report

Overview

- Treat every acute focal neurologic deficit or sudden unexplained neurologic change as a time-critical stroke until proven otherwise.
- Use the Cincinnati Prehospital Stroke Scale for recognition and C-STAT for stroke severity and suspected large-vessel occlusion.
- Record the exact last-known-well time, symptom-discovery time, baseline neurologic function, and witness contact information.

Provider Actions

Recognize stroke, determine timing and severity, protect the airway, and transport rapidly.

EMT

- Perform the Cincinnati Prehospital Stroke Scale and document facial droop, arm drift, and speech findings.
- Complete and document C-STAT when stroke is suspected. Check glucose and treat hypoglycemia under UP-4.
- Determine exact last known well, symptom-discovery time, baseline function, witness contact information, and current anticoagulant or antiplatelet use.
- Keep the patient NPO, protect the affected side, manage secretions, provide oxygen only for hypoxemia, and ventilate with BVM when indicated.
- Activate the stroke alert and begin transport according to the current Claiborne Stroke Destination Plan.

AEMT

- Perform all EMT care. Establish one IV when it can be accomplished without delaying transport.
- Administer normal saline at TKO. Give fluid only for hypotension and reassess for pulmonary edema or volume overload.
- Apply continuous cardiac monitoring when available and support serial vital signs and neurologic examinations.

Paramedic

- Perform all prior care. Obtain a 12-lead ECG when it will not delay transport, particularly when atrial fibrillation or another cardiac cause is suspected.
- Manage airway or ventilation failure and use waveform $EtCO_2$ whenever ventilation is impaired, assisted, or an advanced airway is placed.
- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Repeat an extreme blood pressure manually. For undifferentiated suspected CVA with SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes; reassess every 5 minutes and repeat 10 mg once after at least 10 minutes only if still above the target. Maximum total dose is 20 mg. Target SBP is 220 mm Hg or less and DBP is 110 mm Hg or less. Do not routinely lower blood pressure in the field.
- Coordinate early destination notification and transmit the complete stroke report, including C-STAT score and anticoagulant or antiplatelet last-dose information.

C-STAT Stroke Severity

Maximum score 4; a score of 2 or greater suggests possible large-vessel occlusion.

EMT

- Conjugate gaze deviation: 2 points.
- Incorrectly answers at least one orientation question - age or current month - and fails at least one command - close eyes or open and close hand: 1 point.
- Cannot hold either arm up for 10 seconds: 1 point.
- Document the individual findings and total C-STAT score. A low score does not exclude stroke or eliminate the need for stroke-system transport.

AEMT

- Confirm the documented C-STAT findings during reassessment and report any improvement or deterioration.

Paramedic

- Use C-STAT severity with the current Claiborne Stroke Destination Plan and estimated transport time; do not use the score alone to exclude a patient from stroke activation.

Medication and Bleeding-Risk History

Identify the exact medication, prescribed dose, and last dose time when known.

EMT

- Anticoagulants: warfarin (Coumadin), apixaban (Eliquis), rivaroxaban (Xarelto), dabigatran (Pradaxa), edoxaban (Savaysa), enoxaparin (Lovenox), and heparin injections or infusion.
- Antiplatelet medications: aspirin, clopidogrel (Plavix), ticagrelor (Brilinta), prasugrel (Effient), and aspirin/dipyridamole (Aggrenox).
- Bring medication containers or an accurate medication list when available, but do not delay transport.

AEMT

- Confirm medication name, dose, indication, exact last dose time, missed doses, and any recent bleeding, surgery, trauma, or anticoagulant reversal.

Paramedic

- Include anticoagulant and antiplatelet details in the stroke alert and receiving-facility report because they may affect thrombolysis and hemorrhage treatment decisions.

Indications

- Acute facial droop, unilateral weakness or numbness, speech or language disturbance, gaze deviation, visual loss, or another focal neurologic deficit.
- Sudden unexplained ataxia, severe vertigo, diplopia, dysarthria, unilateral hearing loss, persistent vomiting, or other possible posterior-circulation symptom.
- Resolved focal neurologic symptoms or suspected transient ischemic attack.
- Pediatric patient with a sudden focal neurologic deficit or unexplained acute neurologic change.

Contraindications

- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order.

Assessment | Time and Baseline

- Record exact last known well, exact symptom-discovery time, and the time EMS first assessed the patient.
- For a wake-up stroke, last known well is the last time the patient was observed neurologically normal - not the time symptoms were discovered.
- Document baseline speech, cognition, mobility, dominant hand, functional status, and preexisting neurologic deficits.
- Obtain the witness or caregiver name and telephone number. Bring the witness when practical without delaying transport.
- Do not exclude stroke activation solely because the last-known-well time exceeds 6 hours or is unknown.

Assessment | Neurologic Examination

- Perform and document the Cincinnati screen and C-STAT. Repeat after glucose correction and with any clinical change.
- Assess pupils, gaze, facial symmetry, speech, language, arm and leg strength, sensation, coordination, gait only when safe, and level of consciousness.
- A negative Cincinnati screen does not exclude posterior-circulation stroke. Evaluate sudden severe dizziness or vertigo, ataxia, diplopia, dysarthria, visual loss, unilateral hearing loss, and persistent vomiting.
- Consider seizure or Todd paralysis, hypoglycemia, migraine, intoxication or overdose, trauma, infection, and other stroke mimics without delaying stroke-system transport for persistent focal deficits.

Assessment | Medication and Bleeding History

- Document all current medications, with special attention to anticoagulants and antiplatelet medications.
- For warfarin, apixaban, rivaroxaban, dabigatran, edoxaban, enoxaparin, or heparin, record the prescribed dose and exact last dose time when known.
- For aspirin, clopidogrel, ticagrelor, prasugrel, or aspirin/dipyridamole, record the prescribed dose and exact last dose time when known.
- Ask about prior intracranial hemorrhage, recent surgery or invasive procedure, recent trauma, active bleeding, known bleeding disorder, and recent anticoagulant reversal.

Assessment | Monitoring

- Obtain glucose and at least two complete sets of vital signs when feasible without delaying transport.
- Use continuous cardiac monitoring and obtain a 12-lead ECG when it will not delay transport.
- Use continuous SpO_2. Apply waveform $EtCO_2$ whenever ventilation is impaired or assisted or an advanced airway is placed.

- Repeat the neurologic examination and C-STAT during transport and after every significant clinical change.

Treatment steps

- Support airway and ventilation, suction as needed, and keep the patient NPO. Elevate the head approximately 30 degrees when tolerated and when hypotension or another contraindication is absent.
- Provide oxygen only for hypoxemia and titrate to maintain SpO_2 at or above 94%. Avoid unnecessary hyperoxia.
- Check glucose immediately. Treat confirmed hypoglycemia under UP-4 and repeat the neurologic examination; persistent focal findings remain a stroke alert.
- Activate the stroke alert early and transport according to the current Claiborne Stroke Destination Plan, considering C-STAT severity and estimated transport time.
- Target a scene time of 15 minutes or less. Do not delay transport for nonessential IV attempts, ECG acquisition, blood collection, or an exhaustive mimic investigation.
- Establish one IV when feasible without delaying transport. Use normal saline at TKO and give fluid only for hypotension.
- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Do not routinely lower blood pressure in suspected stroke. For undifferentiated suspected CVA with manually confirmed SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes. Reassess BP, HR, rhythm, perfusion, and neurologic status every 5 minutes; repeat 10 mg once after at least 10 minutes only if still above target. Maximum total dose is 20 mg. Target SBP is 220 mm Hg or less and DBP is 110 mm Hg or less. Do not use this target after thrombolysis; follow AM-06 and written facility orders.
- Transmit glucose, exact last known well, symptom-discovery time, Cincinnati findings, individual C-STAT findings and total score, baseline function, anticoagulant and antiplatelet use with last dose, and witness contact information.
- Transport patients with resolved symptoms or suspected transient ischemic attack because symptom resolution does not eliminate the risk of completed or recurrent stroke.

Medications

Labetalol

Adult age ≥16 only. For undifferentiated suspected CVA with SBP >220 or DBP >110: 10 mg slow IV over 2 minutes; repeat 10 mg once after at least 10 minutes only if still above target; maximum total 20 mg

- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Paramedic standing order after manual confirmation.
- Goal BP is SBP ≤220 and DBP ≤110; do not use this target after thrombolysis or solely to normalize BP.

Warnings / precautions

- Do not administer aspirin, an anticoagulant, or a thrombolytic for undifferentiated suspected stroke. Do not administer an antihypertensive solely to normalize blood pressure.
- Do not administer nitroglycerin solely to lower blood pressure in suspected stroke.
- Naloxone is indicated only when opioid exposure and clinically significant respiratory depression are suspected; do not use it empirically for an isolated focal neurologic deficit.
- Do not delay transport for nonessential procedures, a second IV, blood collection, or complete resolution of diagnostic uncertainty.
- A normal Cincinnati screen, low C-STAT score, symptom improvement, young age, or last-known-well time beyond 6 hours does not exclude a treatment-eligible stroke.

Clinical pearls

- Time is brain, but modern stroke treatment decisions are not limited to the traditional 6-hour window; accurate timing and rapid transport remain essential.
- C-STAT is a severity and destination-support tool, not a rule-out test for stroke.
- Posterior-circulation stroke may present without facial droop or arm weakness and may be missed by common anterior-circulation screens.
- Hypoglycemia may mimic stroke. Correct the glucose abnormality and repeat the examination, but continue the stroke alert when focal deficits persist.
- Avoid hypotension. Routine prehospital blood-pressure reduction may reduce cerebral perfusion and should not be performed for uncomplicated suspected stroke.
- Medication names alone are insufficient: the exact last dose time of an anticoagulant may directly affect receiving-facility treatment.

Special populations | Pediatrics

- Use pediatric protocols for patients younger than 16 years and dose any required supportive medication by actual or length-based weight.
- Stroke occurs in children. Sudden focal weakness, facial asymmetry, speech change, seizure with persistent focal deficit, severe headache, or unexplained acute neurologic change requires stroke-capable evaluation.
- Do not delay transport while attempting to prove an alternative pediatric diagnosis.

Special populations | Posterior-circulation stroke

- Maintain suspicion with sudden severe vertigo or dizziness, inability to walk, marked ataxia, diplopia, dysarthria, visual loss, unilateral hearing loss, persistent vomiting, or crossed neurologic findings.
- A negative Cincinnati screen does not exclude posterior stroke. Activate the stroke system when the overall examination and onset are concerning.

Source and references

Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025 - SOP 304 CVA / Stroke. | American Heart Association / American Stroke Association - 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke. | McMullan et al. - Prospective Prehospital Evaluation of the Cincinnati Stroke Triage Assessment Tool, Prehospital Emergency Care, 2017. | Claiborne EMS UP-4 Diabetic / Glucose Emergencies protocol. | North Carolina College of Emergency Physicians UP-14 Suspected Stroke source protocol retained for historical comparison.

Revision: October 31, 2026 | Last verified: October 31, 2026